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RAMAIAH Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END EXAMINATIONS – APRIL 2023

Program	:	**B.E. - CSE (Cyber Security) / CSE (Artificial Intelligence and Machine Learning)**	**Semester**	:	**III**
Course Name	:	**Data Structures**	**Max. Marks**	:	**100**
Course Code	:	**CY33 / CI33**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	Differentiate between static and dynamic memory allocation. With an example code snippet illustrate allocating memory dynamically to store array of 10 integer elements.	CO1	(10)
	b)	Differentiate between structures and unions. Write a C program to create student database using structures in C to store various information about student such as name, USN, branch, marks scored by student in 3 tests. Calculate the average of best two test marks scored by each student. Display the details of student who has scored highest average marks in CSE department.	CO1	(10)
2.	a)	Write a c program to multiply the elements of 2 matrix using pointers.	CO1	(08)
	b)	What is sparse matrix? Explain the process of representing sparse matrix using array data structure.	CO1	(08)
	c)	What is the significance of malloc() and calloc()? Write the program to illustrate the operation of malloc().	CO1	(04)

UNIT - II

3.	a)	Demonstrate the stack operations during the conversion of the following infix expression to prefix expression.((M * N)*O+(P /Q)).	CO2	(08)
	b)	Write a program to find x^n using recursion.	CO2	(06)
	c)	Explain different types of recursion with an example for each.	CO2	(06)
4.	a)	Develop C functions to perform PUSH and POP operations on an integer stack. Write necessary statements to check for stack overflow and stack underflow conditions.	CO2	(06)
	b)	Develop a recursive C function to find n th Fibonacci number. Using this function, develop a C program to generate Fibonacci series.	CO2	(06)
	c)	Develop a C program to evaluate a valid postfix expression.	CO2	(08)

UNIT - III

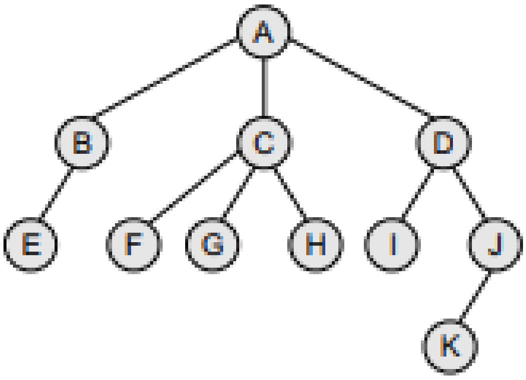
5.	a)	Represent two queues in an array, *memory*[MEMORY_SIZE]. Write C functions to perform add and delete operations on these queues. Add function should be able to add elements to the queues as long as the total number of elements in both queues is less than MEMORY_SIZE -1.	CO3	(06)
	b)	Write C functions to perform add and delete operations on a circular queue using dynamic implementation.	CO3	(08)
	c)	Describe limitations of a linear queue with suitable examples.	CO3	(06)
6.	a)	Explain implementation of different operations on dequeue.	CO3	(06)
	b)	Write a C function to compare contents of two circular queues using array implementation.	CO3	(08)
	c)	Illustrate the applications of queues with an example.	CO3	(06)

UNIT- IV

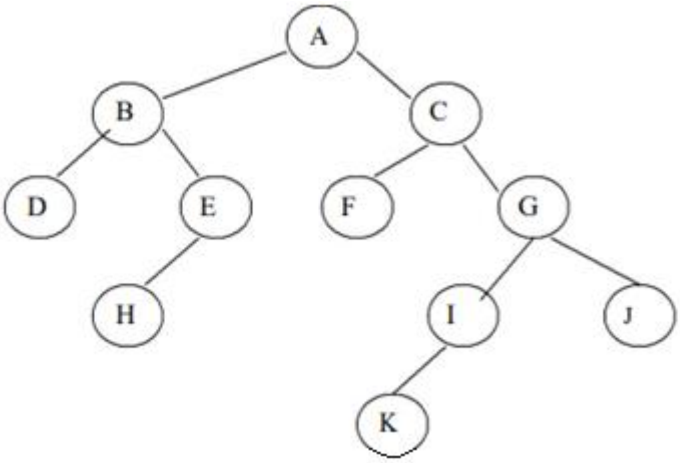
7.	a)	Briefly discuss how to implement insertion and deletion operations on Queue using linked lists by taking an example.	CO4	(10)
	b)	Write the C Program to Delete the Node Before and after a Given Node in a Doubly Linked List.	CO4	(10)
8.	a)	Write a C program for Deleting the First and Last Node from a Circular Linked List.	CO4	(10)
	b)	Illustrate the process of inserting the new node in a doubly linked list for different cases.	CO4	(10)

UNIT - V

9.	a)	Explain different steps involved to convert an postfix expression into an binary expression tree and Draw the binary expression tree that represents the following postfix expression: A B + C * D –	CO5	(08)
	b)	Illustrate different steps for creating binary tree from a general tree given below:	CO5	(08)



	c)	Discuss the properties of Binary tree.	CO5	(04)
10.	a)	Illustrate the process of traversing the given binary tree using Level order, inorder, preorder and post order traversal technique and write the output.	CO5	(08)



	b)	Explain the Linked List and Array representation of binary trees by taking an example.	CO5	(08)
	c)	Discuss the characteristics of complete and extended binary tree with an example for each.	CO5	(04)

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